



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

CyberSecurity

0 - “Introduction to the Course”

Gabriele D'Angelo <g.dangelo@unibo.it>

<https://www.unibo.it/sitoweb/g.dangelo/en>



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Cyber(in)Security

0 - “Introduction to the Course” *“Small introduction to the Paranoia”*

Gabriele D'Angelo <g.dangelo@unibo.it>

<https://www.unibo.it/sitoweb/g.dangelo/en>



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CyberSecurity: */me*

Instructor:

- **Gabriele D'Angelo** <g.dangelo@unibo.it>
- Webpage: <https://www.unibo.it/sitoweb/g.dangelo/en>
- LinkedIn: <https://www.linkedin.com/in/gdangelo/>



Office hours:

- by appointment (e-mail)



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CyberSecurity: *teaching tutor*

Tutor:

- **Manuel Andruccioli** <manuel.andruccioli@unibo.it>
- Webpage: <https://www.unibo.it/sitoweb/manuel.andruccioli/en>

Office hours:

- by appointment (e-mail)



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CyberSecurity: *course material*

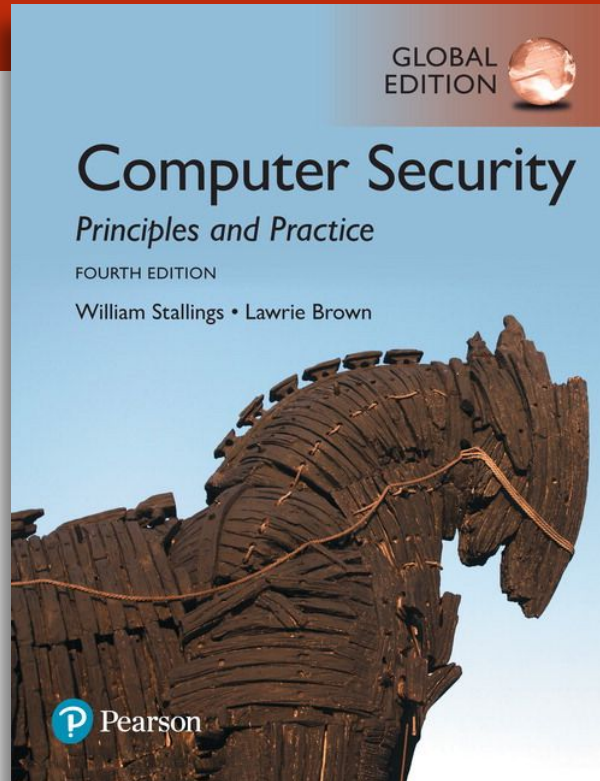
Available on **Virtuale** ([link](#)):

- all the teaching material used during the lessons
- the “**Forums**” will be used to report the content of every lesson and to provide further info about the course
- **please read the FAQ**



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CyberSecurity: *book*



Computer Security: Principles and Practice

William Stallings, Lawrie Brown

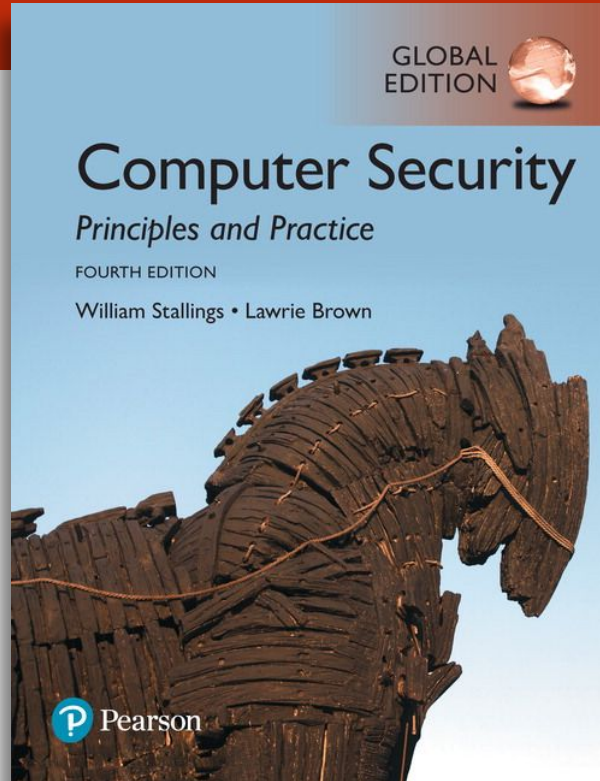
(forth) global edition

Pearson



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CyberSecurity: *book*



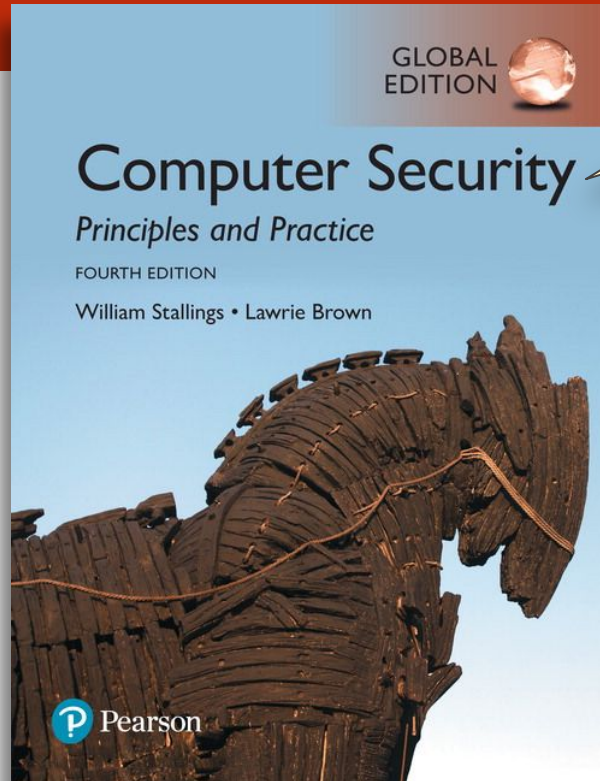
WARNING:

The goal of the slides is to
INTEGRATE the book. Studying
the slides is ***not* sufficient** for a
decent knowledge of the topics



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CyberSecurity: *book*



In this document you find
what parts can be **skipped**
vs. must be studied

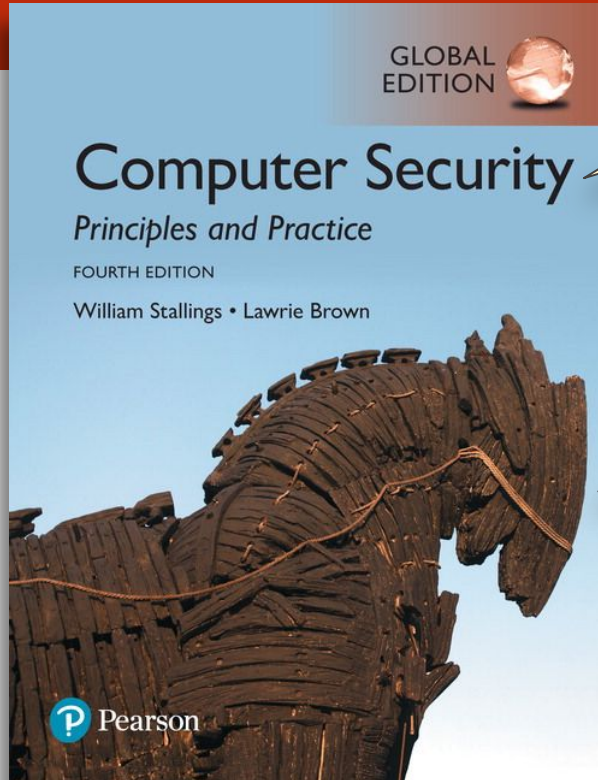
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CyberSecurity: *book*



In [this document](#) you find
what parts can be **skipped**
vs. must be studied

The core of the course is
based on the Chapters from

1 up to 4

plus some **optional**
integrations in other chapters



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CyberSecurity: *lessons*

Timetable:

- Tuesday, 09-11, classroom “3.4 - Aula Magna”
- Thursday, 14-17, classroom “2.12”

Modules:

- Module 1: “**CyberSecurity**” (/me)
- Module 2: “**CyberSecurity of AI**” (Prof. Stefano Ferretti)



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CyberSecurity: *lessons*

Timetable:

- Tuesday, 09-11, classroom
- Thursday, 14-17, classroom

Modules:

- Module 1: “**CyberSecurity**” (Prof. Stefano Ferretti)
- Module 2: “**CyberSecurity of AI**” (Prof. Stefano Ferretti)

TOPICS:

- cybersecurity of **ML models**
- **malicious** activity detection
- systems for **data protection**



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CyberSecurity: *lessons*

Timetable changes:

- reported ASAP on **Virtuale** + **email notification**

Theory vs. Laboratory

- there is no specific laboratory (not enough time) but we will see also **lab examples** during the lessons



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CyberSecurity: *degree programs*

93470 - Cybersecurity - 6 cfu

Campus: Cesena

Degree Second cycle degree programme (LM) in Computer Science and Engineering

programme: Second cycle degree programme (LM) in Digital Transformation Management

Lesson period: from February 17, 2026 to May 14, 2026

 [Course timetable](#) |  [Teaching resources on Virtuale](#)

DTM



LM_ISI



+



Digital



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CyberSecurity: *students assessment*

Final examination = written test at the PC (in English)

- **5 open-ended questions** (max 6 points for each question)
- **1 hour and 30 minutes**
- closed book, no written notes, no collaboration etc.
- **questions on:** slides + book + mandatory readings + tutorials
- **no questions on:** news



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CyberSecurity: students assessment

Final examination

MANDATORY READINGS

Some documents (i.e., papers / articles) about specific cybersecurity topics

“Read at home” and then “discuss in the classroom”

- 5 open-
- 1 hour and
- closed book, no written notes, no collaboration, etc.
- **questions on:** slides + book + mandatory readings + tutorials
- **no questions on:** news



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CyberSecurity: *students assessment*

Final examination = written

- **5 open-ended questions**
- **1 hour and 30 minutes**
- closed book, no written notes, no collaboration etc.
- **questions on:** slides + book + mandatory readings + tutorials
- **no questions on:** news

TUTORIALS

Documents (prepared by me) about
practical examples / lab activity



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CyberSecurity: *students assessment*

NEWS FROM THE “REAL WORLD”

Most lessons will start **presenting** (and then **discussing together**) what just happened in the cybersecurity field (at large)

Final examination

- 5 open-
- 1 hour and 30 min
- closed book, no whiteboards, no collaboration etc.
- **questions on:** slides + book + mandatory readings + tutorials
- **no questions on:** news



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CyberSecurity: *students assessment*

Final examination = written test at the PC (in English)

- **5 open-ended questions** (max 6 points for each question)
 - **4 questions on the course program**
 - **3** about “Module 1” + **1** about “Module 2”
 - **1 question on the mandatory readings**



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CyberSecurity

QUESTIONS

Different questions for different degree programs:

DTM vs. LM_ISI (+ others)

Final examination

- **5 open-ended questions** (max 6 points for each question)
 - **4** questions on the **course program**
 - **1** question on the **mandatory readings**



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CyberSe

GENERAL RULE

Detailed description of the **most technical aspects** is
suggested in DTM and ***mandatory*** in LM_ISI

Final examination

- **5 open-ended questions** (max 6 points for each question)
 - 4 questions on the **course program**
 - 1 question on the **mandatory readings**



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CyberSe

Final examination

GENERAL RULE

Detailed description of the **digital transformation aspects** (e.g., economic issues) is ***mandatory*** in DTM and ***suggested*** in LM_ISI

- **5 open-ended questions** (max 6 points for each question)
 - **4 questions on the course program**
 - **1 question on the mandatory readings**



“INVARIANT” QUESTION EXAMPLE

«What is the “CIA Triad”? Describe in detail all the components of the Triad and their importance for computer security»

- **5 open-ended questions** (max 6 points for each question)
 - **4** questions on the **course program**
 - **1** question on the **mandatory readings**



SPECIFIC QUESTION EXAMPLE: “DTM version”

«What is a pharming attack?»

- **5 open-ended questions** (max 6 points for each question)
 - 4 questions on the **course program**
 - 1 question on the **mandatory readings**



SPECIFIC QUESTION EXAMPLE: “LM_ISI version”

«What is a pharming attack? Describe all the protocols that are typically involved in this attack»

- **5 open-ended questions** (max 6 points for each question)
 - **4** questions on the **course program**
 - **1** question on the **mandatory readings**



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CyberSecurity: *students assessment*

Final examination = written test at the PC (in English)

- **5 open-ended questions** (max 6 points for each question)
 - **4 questions on the course program**
 - **1 question on the mandatory readings**

**Some other examples
of questions can be
found on [Virtuale](#)**



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CyberSecurity: *students assessment*

MANDATORY READINGS

Find

It is required to carefully ***SUMMARIZE*** the paper **AND** then to discuss it based on ***your*** **OPINIONS** (i.e., **critical thinking**) and **KNOWLEDGE**

- 4 questions on the **course program**
- 1 question on the **mandatory readings**



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CyberSecurity: *students assessment*

Evaluation methodology

- The evaluation is based on **different aspects**
- More details can be found in [this document](#)
- **Grades** must be **refused** within **5 days** from the notification otherwise they are considered as **accepted** and **recorded**



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CyberSecurity: *exam sessions*

High level schedule:

- **3** sessions between **June** and **July**
- **1** session in **September**
- **2** sessions between **January** and **February**



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CyberSecurity: *exam sessions*

Next sessions (please check on [AlmaEsami](#)):

- **June**

- **16/06/2026** (Lab. 2.2 “Vela”)
- **02/07/2026** (Lab. 2.2 “Vela”)

- **July**

- **21/07/2026** (Lab. 2.2 “Vela”)



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CyberSecurity: *exam sessions*

Next sessions (please check on [AlmaEsami](#)):

- **June**

- 16/06/2024
- 02/07/2024

- **July**

- 21/07/2024

WARNING:

Multiple turns will be needed!

Check about your specific turn!

CyberSecurity: *main topics* (Module 1)



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- **Introduction** to computer security
- Overview: main **concepts** and **terminology**
- Security of the **Internet-of-Things**
- **Denial-of-Service (DoS)** attacks
- **Case studies** (a lot of)
- **Network security**: exposed attack surface (i.e., nmap)

CyberSecurity: *main topics* (Module 1)



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- Basic **cryptographic** concepts (+ **end-to-end encryption**)
- Attacking the **encryption** + **pharming** attack (i.e., http vs. https)
- User **authentication**
- **Password cracking** (i.e., John the Ripper, GPU cracking)
- **Access control**
- **Reading assignments** (mandatory)
- Security-related **news** (report and discussion)

CyberSecurity: *main topics* (Module 1)



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- **EU legislation** on CyberSecurity
- CyberSecurity and **AI**
- **Hardening** of Linux systems
- The TOR network (i.e., **anonymous** communications)
- **Cryptocurrencies** and **blockchains**
- **Operational** security (OPSEC)
- Security of **mobile devices**

MSc Thesis with my research group?



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- CyberSecurity
- CyberSecurity and AI
- Simulation techniques
- Complex systems
- Blockchains
- ... and a mix of them

The screenshot shows the website for the 'ANALYSIS OF NETWORKS AND SIMULATION RESEARCH GROUP'. The header includes the Alma Mater Studiorum University of Bologna logo and the group name. A navigation bar has links for HOME, PEOPLE, PUBLICATIONS, and AGENDA. The main content area features a 'Highlights' section with three cards: 'Group Description' (showing a red carpet with 'ALMA' written on it), 'Research Topics' (showing mathematical notation $\langle a \rangle$, $[a]$, and $\langle b \rangle$), and 'Publications' (showing a grid of yellow book covers). At the bottom, there are links to the 'GITHUB' page (with a code icon) and the 'PADS RESEARCH GROUP' (with a brain icon and text: 'Parallel and Distributed Simulation (PADS) Research Group').

CyberSecurity: *why?*



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WHY?

CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

By **Sergiu Gatlan**

September 21, 2023 01:57 PM 1



Apple released emergency security updates to patch three new zero-day vulnerabilities exploited in attacks targeting iPhone and Mac users, for a total of 16 zero-days fixed this year.



CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

By [Sergiu Gatlan](#)

September 21, 2023 01:57 PM 1



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Vulnerability

CyberSecurity: *vulnerability*



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«In cybersecurity, **vulnerability** refers to a **weakness** or **flaw** in a system's **design**, **implementation**, or **configuration** that could be exploited by **threat actors** to compromise the confidentiality, integrity, or availability of the system or its data. Vulnerabilities can exist in various components of a computer system, including **software**, **hardware**, **network protocols**, and **human factors**». (ChatGPT 3.5, 04/03/2024)

CyberSecurity: **zero-days**



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September 21, 2023 01:57 PM 1



September 21, 2023

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CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

By [Sergiu Gatlan](#)

September 24, 2015 01:57 PM 1



Apple released emergency security updates to patch three new zero-day vulnerabilities exploited in attacks targeting iPhone and Mac users, for a total of 16 zero-days fixed this year.

“zero-day”

CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

By **Sergiu Gatlan**

September 21, 2023 01:57 PM 1



Apple released emergency security updates to patch three new zero-day vulnerabilities exploited in attacks targeting iPhone and Mac users, for a total of 16 zero-days fixed this year.

Zero-day: “**vulnerability** in a computer system that was **previously unknown** to its developers or anyone **capable of mitigating it**”

CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

By [Sergiu Gatlan](#)

September 21, 2023 01:57 PM 1



“exploited”

Apple released emergency security updates to patch three new zero-day vulnerabilities exploited in attacks targeting iPhone and Mac users, for a total of 16 zero-days fixed this year.

CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

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**“16 zero-days fixed
in 2023”**

Apple released emergency security updates to patch three new zero-day vulnerabilities exploited in attacks targeting iPhone and Mac users, for a total of 16 zero-days fixed this year.

CyberSecurity: **zero-days**



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Apple emergency updates fix 3 new zero-days exploited in attacks

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September 21, 2023 01:57 PM 1



**“16 zero-days fixed
in 2023”**

Apple released emergency security updates to patch three new zero-day vulnerabilities exploited in attacks targeting iPhone and Mac users, for a total of 16 zero-days fixed this year.

Google Project Zero

VULNERABLE DEVICES

CyberSecurity: *healthcare*



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U.S. Department of Health and Human Services



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[ice Safety](#) > [Safety Communications](#)

Cybersecurity Vulnerabilities of Hospira Symbiq Infusion System: FDA Safety Communication



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LINKEDIN



PIN IT



EMAIL



PRINT

Date Issued: July 31, 2015

Audience: Health care facilities using the Hospira Symbiq Infusion System

Device: Symbiq Infusion System, Version 3.13 and prior versions

The Hospira Symbiq Infusion System is a computerized pump designed for the continuous delivery of general infusion therapy for a broad patient population.

CyberSecurity: *healthcare*



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U.S. Department of Health and Human Services



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[Device Safety](#) > [Safety Communications](#)

Cybersecurity Vulnerabilities of Hospira Symbiq Infusion System: FDA Safety Communication



SHARE



TWEET



LINKEDIN



PIN IT



EMAIL



PRINT

Date Issued: July 31, 2015

Audience: Health care facilities using the

System

Device: Symbiq Infusion System, Version

2015

The Hospira Symbiq Infusion System is a... designed for the continuous delivery of general infusion therapy for a broad patient population.

CyberSecurity: *healthcare*



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Problem solved?



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SCIENCE HEALTH

Medtronic issues 'urgent' recall of insulin pump controller vulnerable to hacks

It's not the first alert around these devices

By Nicole Wetman | Oct 6, 2021, 1:00pm EDT

f t SHARE

2021

A photograph showing several stacks of Medtronic insulin pump boxes. The boxes are white and orange, with labels that include QR codes and text such as "Medtronic", "Insulin Pump", and "Controller". The boxes are stacked in a way that shows multiple units of the device.

CyberSecurity: *healthcare*



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SCIENCE HEALTH

Medtronic issues 'urgent' recall of insulin pump controller vulnerable to hacks

It's not the first alert around these devices

By Nicole Wetsman | Oct 6, 2021, 1:00pm EDT



SHARE



“2021”



MEDTECH

BD warns of 8 cybersecurity flaws in Alaris infusion pump systems

By Andrea Park · Jul 14, 2023 1:15pm

BD

Cybersecurity

Infusion pump

hacking



“2023”

CyberSecurity: *healthcare*



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FDA

“2024”

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IN THIS SECTION: Medical Device Recalls

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Infusion Pump Software Correction: Fresenius Kabi USA, LLC, Issues Correction for Ivenix Infusion System Large Volume Pump (LVP) Software due to Multiple Anomalies that May Cause Delay or Underdosage of Therapy

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*This **recall** involves correcting devices, and does not involve removing them from where they are used or sold. The FDA has identified this recall as the most serious type. This device may cause serious injury or death if you continue to use it without correction.*



CyberSecurity: *healthcare*



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A screenshot of a website article from the Foundation for Defense of Democracies (FDD). The article is dated July 9, 2025, and is a Policy Brief. The title is "Medical Devices Still Vulnerable to Hacking Despite New FDA Guidance". The background image shows the FDA seal and the text "Department of Health and Human Services". A yellow speech bubble with the text "2025" points to the date. The FDD logo is in the top left, and navigation links (About, Issues, Projects, Prod) are in the top right. A search icon is also present. At the bottom, there are social media icons and a breadcrumb trail "Home » Cyber".

FDD
FOUNDATION FOR
DEFENSE OF DEMOCRACIES

About Issues Projects Prod

“2025”

July 9, 2025 | Policy Brief

Medical Devices Still Vulnerable to Hacking Despite New FDA Guidance

Department of Health and Human Services

Home » Cyber

✕ f e p



Samantha Ravich
CCTI Chairman



Johanna Yang
CCTI Research and Editorial Associate

CyberSecurity: *healthcare*



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SCIENCE / HEALTH

Cyberattack delays patient care at major US hospital chain



Doctors rely on computer systems to deliver care. Getty

/ The chain operates over 700 healthcare sites around the country

By **NICOLE WETSMAN**

Oct 11, 2022, 7:26 PM GMT+2 | [2 Comments](#) / [2 New](#)



Patients at hospitals in one of the largest healthcare systems in the United States are facing delays in care due to a cyberattack. The disruption is stretching into its second week.



VULNERABLE NETWORKS

CyberSecurity: *Denial of Service*



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Massive DDoS Attack

Spotify, Twitter, Github, Etsy, and More Go Offline

CyberSecurity: *Denial of Service*



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21/10/2016 – 1.2

Tbps

Mirai IoT botnet

Massive DDoS Attack

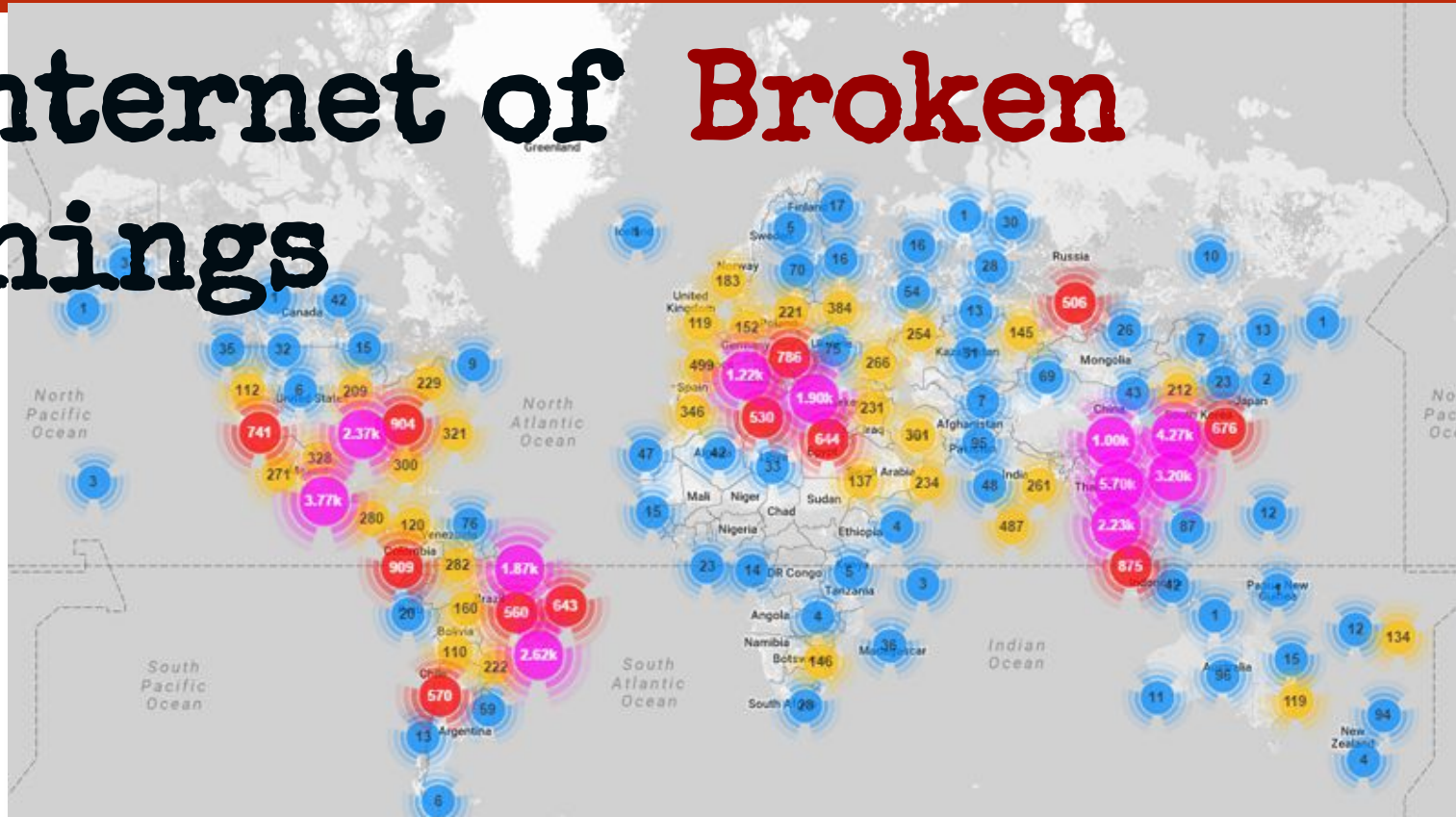
Spotify, Twitter, Github, Etsy, and More Go Offline

CyberSecurity: *Denial of Service*



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Internet of Broken Things



CyberSecurity: *Denial of Service*



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Famous DDoS attacks: The largest DDoS attacks of all time

In a distributed denial-of-service (DDoS) attack, multiple devices are used to overwhelm a targeted server with traffic and take online services offline. Some of the biggest DDoS attacks have made major tech headlines.

[Learning Center](#)[What is a DDoS attack?](#)[What is a DDoS botnet?](#)[Common DDoS attacks](#)[Flood attacks](#)[DDoS attack tools](#)[Glossary](#)[th](#)

INSECURE BY DESIGN

CyberSecurity: *Internet*



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Internet security saviour Daniel Kaminsky dies at 42

Researcher was known for his discovery of a fundamental flaw in the fabric of the internet

© Tue, Apr 27, 2021, 11:21

Nicole Perloth



Daniel Kaminsky found a serious flaw in the internet's basic plumbing that could allow skilled coders to take over websites, siphon off bank credentials or even shut down the internet.



Daniel Kaminsky, a security researcher known for his discovery of a fundamental flaw in the fabric of the internet, died Friday at his home in San Francisco. He was 42.



«The Internet was never
designed to be secure.
The Internet was designed
to move pictures of cats»

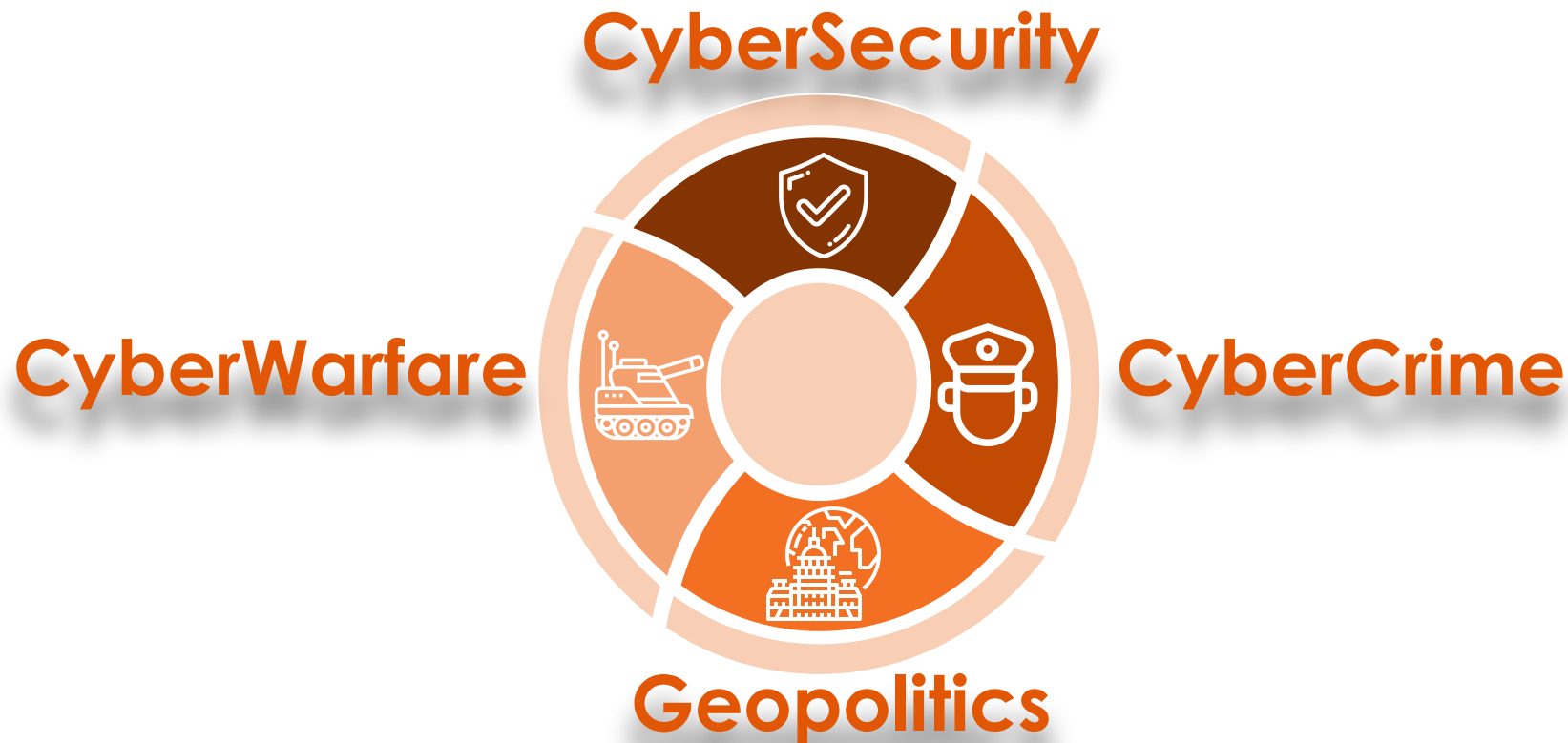
(Dan Kaminsky)

RISK MANAGEMENT

CyberSecurity (*at large*)



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CyberSecurity at large



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One example is
“ransomware”

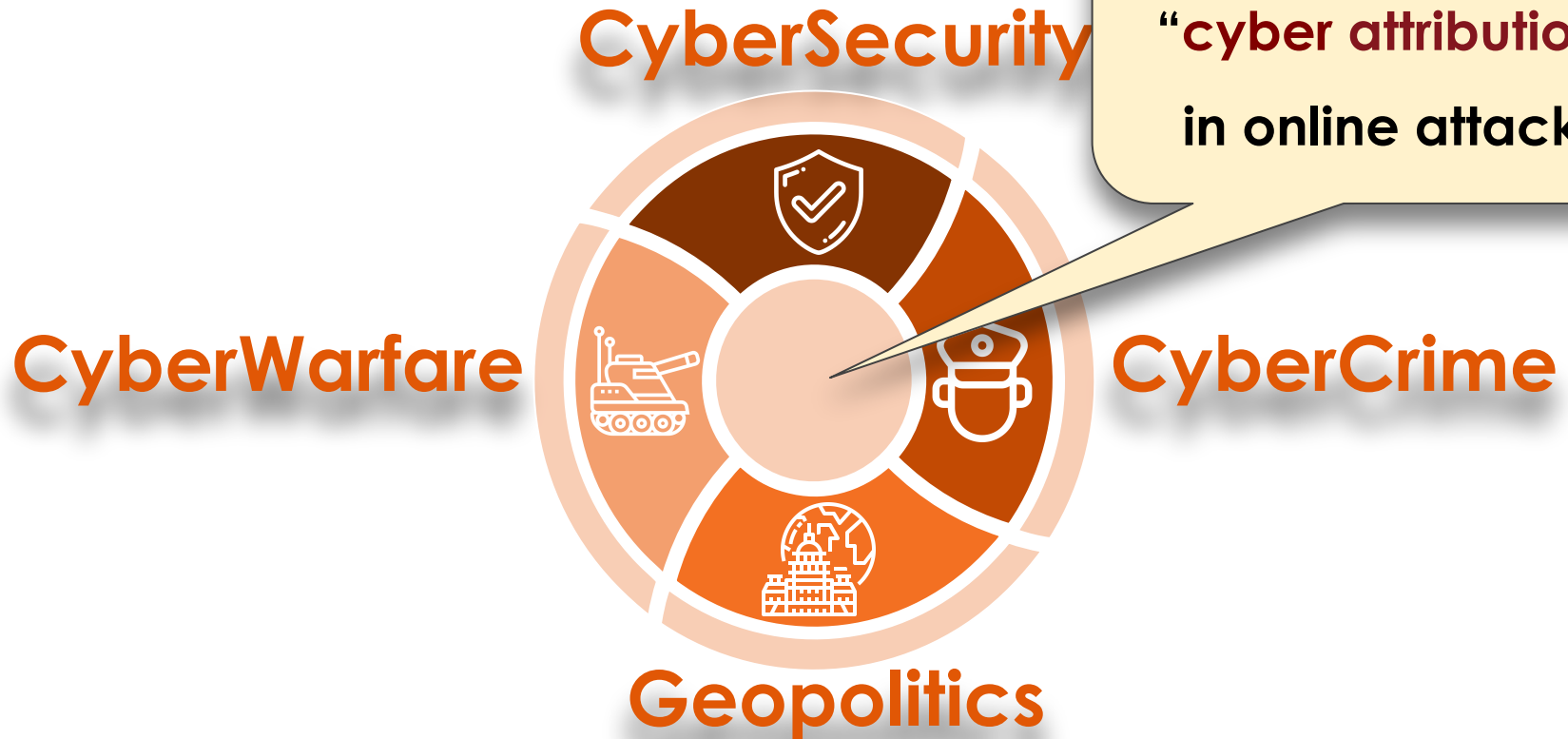
CyberWarfare



CyberCrime

Geopolitics

CyberSecurity at large



CyberSecurity: *risk management*



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[Common exclusions to look out for](#)

[Cyber insurance in action](#)

[How to buy cyber insurance](#)

[Advice for improving cyber security](#)

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Cyber insurance - Common exclusions

As with any insurance policy, it is crucial to review not only what is covered by your insurer but also what is excluded. You should be looking at exclusions and also definitions and conditions when examining your policy. Many exclusions in cyber insurance are the same as those in other insurance policies such as war and terrorism but there are also some that are specific to cyber insurance, including:



CyberSecurity: *risk management*



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Cyber Insurance

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Cyber insurance - Common exclusions

As with any insurance policy, it is crucial to review not only what is covered by your policy but also what is excluded. When examining your policy, you should be looking at exclusions and also definitions and conditions when examining your policy. Cyber insurance are the same as those in other insurance policies such as war and terrorism but with some specific to cyber insurance, including:

A photograph showing three people in a modern office environment. Two men and one woman are gathered around a desk with multiple computer monitors, appearing to be in a collaborative work or discussion.

“Losses to businesses that result from **cyber warfare** and **cyber-attacks** that may be linked to the actions of a particular **country** or **government** are **common exclusions** due to the risks being so large and beyond the capacity of individual insurers”



FIND THE
VALUE

CyberSecurity: *value*



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North Korean hackers increasingly targeting wealthy crypto holders

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Joe Tidy

Cyber correspondent, BBC World Service



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Targeting high net worth crypto holders has helped North Korean hackers steal more than \$2bn (£1.49bn) so far this year according to researchers.



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Where is the value?

There is a lesson to be learned...

Targeting high net worth crypto holders has increased by more than \$2bn (£1.49bn) so far this year





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CyberSecurity

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